Appendix 1 Figures: Predicting density and assessing factors driving marbled murrelet distribution in Puget Sound using hierarchical distance sampling

Code by S.J. Gillman

Required Packages

```
library(tidyverse)

library(MCMCvis)
library(nimble)

library(ggplot2)
library(cowplot)
library(patchwork)
library(sf)
library(viridis)
library(scico)
library(ggtext)
```

Required Files Previously Made or Provided

- `Survey_Grid_Info.RData`
- `chain_1.rds`
- `chain_2.rds`
- `chain_3.rds`
- `nimConstants.rds`
- `nimData.rds`
- `grid2km_hex.gpkg`
- `all_tracks_2026_final.gpkg`
- `predicted_month_year_global.rds`
- `WA_State_Boundary.shp` downloaded from the Washington State Department of Natural Resources GIS Open Data Portal
- `WSDOT_-_Major_Shorelines.shp` also downloaded from the Washington State Department of Natural Resources GIS Open Data Portal

Each of the `CH1_nSQ_results_X.RData` contains the posterior samples, the data and constants that went into the model, and the initial values.

```
load("../DataAnalysis/DataFiles/Survey_Grid_Info.RData")
samples_1 <- readRDS("RESULTS/COMPRESSED/chain_1.rds")
samples_2 <- readRDS("RESULTS/COMPRESSED/chain_2.rds")
samples_3 <- readRDS("RESULTS/COMPRESSED/chain_3.rds")
```

```r
nimConstants <- readRDS("RESULTS/COMPRESSED/nimConstants.rds")
nimData <- readRDS("RESULTS/COMPRESSED/nimData.rds")


grids <- st_read("..CovariateProcessing/Outputs/grid2km_hex.gpkg")

tracklines <- st_read("../ProcessingCovariates/GIS_COVS/all_tracklines_2026_final.gpkg")


tracks <- tracklines %>%
  mutate(numid = row_number()) %>%
  st_buffer(dist = 215)

WA_State <- st_read("WA_State_Boundary/WA_State_Boundary.shp")
puget <- st_read("WSDOTMajor_Shorelines/WSDOT_-_Major_Shorelines.shp")
WA_State <- st_transform(WA_State, crs = st_crs(grids))
puget <- st_transform(puget, crs = st_crs(grids))
```

```r
chains <- coda::mcmc.list(samples_1,samples_2,samples_3)



result_summary <- MCMCsummary(chains, probs = c(0.025,0.25,0.5,0.75, 0.975)) %>%
  rownames_to_column("parameter") %>%
  mutate(
    variable = sub("\\[.*", "", parameter),
    model_id = gsub("\\]", "", sub(".*\\[", "", parameter)))

# posterior direction
pd_dir <- bayestestR::p_direction(chains)
```

Appendix 1

```r
stat_vals <- as.data.frame(t(t(colnames(nimData[["STAT_COVS"]])))) %>%
  rownames_to_column("model_id") %>%
  mutate(V1 = toupper(gsub("_scale", "", V1)))


sea_vals <- as.data.frame(t(t(colnames(nimData[["DYNAMIC"]])))) %>%
  rownames_to_column("model_id") %>%
  mutate(V1 = toupper(gsub("_scale", "", V1)),
         model_id = paste0("1, ", model_id)) %>%
  rbind(as.data.frame(t(t(colnames(nimData[["DYNAMIC"]])))) %>%
          rownames_to_column("model_id") %>%
          mutate(V1 = toupper(gsub("_scale", "", V1)),
                 model_id = paste0("2, ", model_id)))

result_coef <- result_summary %>%
  full_join(pd_dir %>% rename(parameter = Parameter),
            by = "parameter") %>%
  filter(variable == "eps_mon") %>%
```

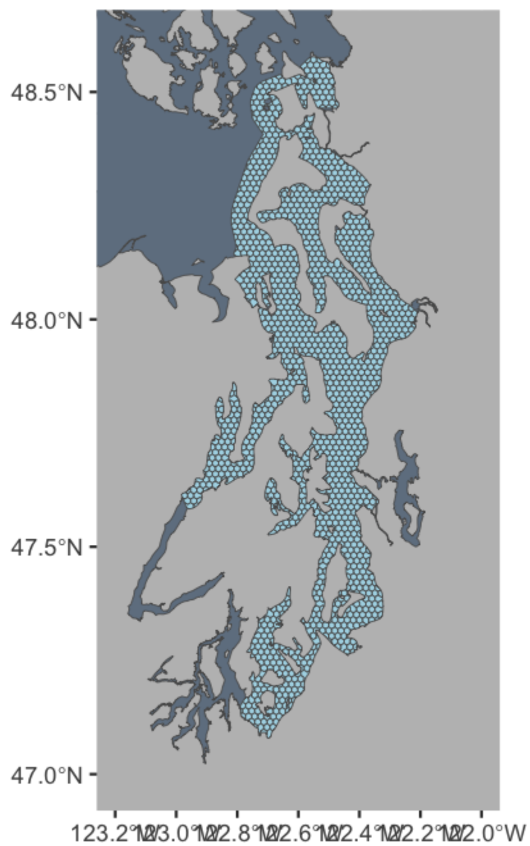
```r
rename(mNum = model_id) %>%
full_join(survey_info %>%
            mutate(mNum = as.character(mNum)) %>%
            distinct(month, mNum), by = "mNum") %>%
mutate(parameter = paste0("eps-", month)) %>%
dplyr::select(-month) %>%
rename(model_id = mNum) %>%
# simple
rbind(result_summary %>%
        full_join(pd_dir %>% rename(parameter = Parameter),
                  by = "parameter") %>%
        filter(variable == "alpha0" |
         variable == "mu_gt" |
         variable == "mu_tide" |
         variable == "mu_delta" |
         variable == "rate_r" |
         variable == "shape_r" |
         variable == "sig_delta" |
         variable == "sig_gt" |
         variable == "sig_pair" |
         variable == "sig_tide" |
         variable == "sig_mon"  |
         variable == "sigma_mean"  |
         startsWith(variable, "beta"))) %>%
# static
rbind(result_summary %>%
        full_join(pd_dir %>% rename(parameter = Parameter),
                  by = "parameter") %>%
        filter(variable == "alpha_stat") %>%
        full_join(stat_vals, by = "model_id") %>%
        mutate(parameter= V1) %>%
        dplyr::select(-V1)) %>%
# seasonal
rbind(result_summary %>%
        full_join(pd_dir %>% rename(parameter = Parameter),
                  by = "parameter") %>%
        filter(variable == "alpha_sea") %>%
        full_join(sea_vals, by = "model_id") %>%
        mutate(parameter= ifelse(startsWith(model_id, "1,"),
                                 paste0(V1, " B"),
                                 paste0(V1, " NB"))) %>%
        dplyr::select(-V1)) %>%
rbind(result_summary %>%
        full_join(pd_dir %>% rename(parameter = Parameter),
                  by = "parameter") %>%
filter(variable == "eps_pair")) %>%
# sigma
mutate(parameter = ifelse(parameter == "beta", "TIDE",
                   ifelse(parameter == "alpha0[1]","ALPHA0 B",
                   ifelse(parameter == "alpha0[2]","ALPHA0 NB",
                   ifelse(variable == "eps_pair",
                          paste0("EPS_PAIR-", model_id),
                          toupper(parameter)))))) %>%
```

```
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  rename(ESS = `n.eff`)


result_est <- result_summary %>%
  full_join(pd_dir %>% rename(parameter = Parameter), by = "parameter") %>%
  # group size & pcap
  filter(variable == "gs.expected" |
           variable == "pcap_mean") %>%
  mutate(myNum = model_id) %>%
  full_join(survey_info %>%
              distinct(myNum, monyear) %>%
              mutate(myNum = as.character(myNum)),
            by = "myNum") %>%
  mutate(parameter = ifelse(variable == "gs.expected",
                            paste0("GS- ", monyear),
                            paste0("PCAP- ", monyear))) %>%
  dplyr::select(-c(myNum, monyear)) %>%
   mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  rename(ESS = `n.eff`) %>%
  dplyr::select(-c(model_id,variable))
```

```
write_csv(result_est, file = "RESULTS/Supplemental/parameter_ests.csv")
```

Figure 1: Grid cell-transect overlap frequency

```
ggplot()+
  geom_sf(data = WA_State, fill ="gray") +
  geom_sf(data=puget, fill="slategray")+
  geom_sf(data = grids, fill = "lightblue") +
  coord_sf(xlim = c(-122, -123.2), ylim = c(47, 48.6), crs = st_crs(4326))
```



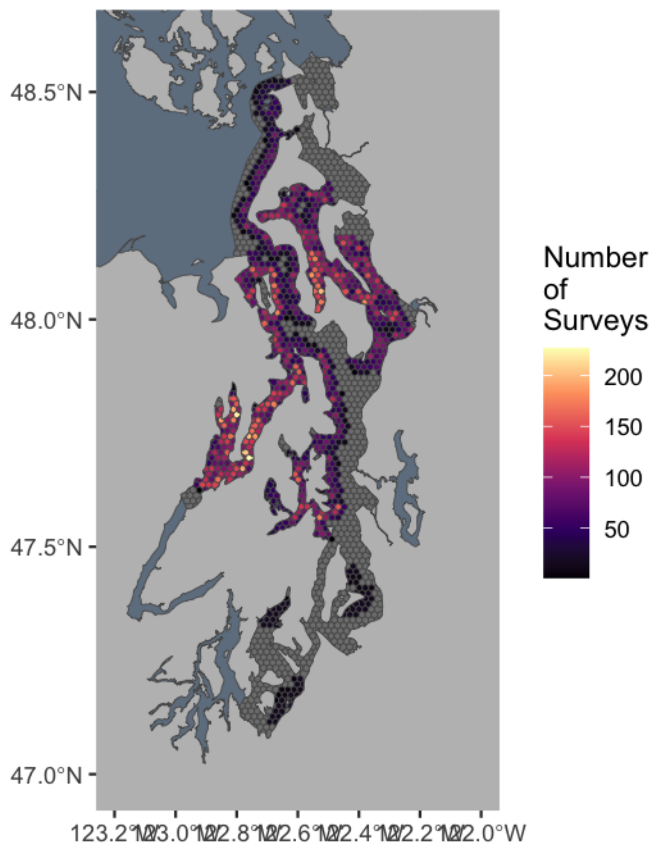

```
hex_overlap <- grids %>%
  st_intersection(tracks %>% select(trip_date))
```

```
## Warning: attribute variables are assumed to be spatially constant throughout
## all geometries
```

```
freq <- grids %>% select(ID) %>%
  full_join(hex_overlap %>%
  st_drop_geometry() %>%
  group_by(ID) %>%
  count() %>%
  ungroup(), by = "ID")

length(which(!is.na(freq$n)))
```

```
## [1] 893
```

```
ggplot()+
  geom_sf(data = WA_State, fill ="gray") +
  geom_sf(data=puget, fill="slategray")+
  geom_sf(data = freq, aes(fill =n)) +
  coord_sf(xlim = c(-122, -123.2), ylim = c(47, 48.6), crs = st_crs(4326))+
  scale_fill_viridis_c(option = "magma")+
  labs(fill = "Number\nof\nSurveys")
```




```r
#ggsave("RESULTS/Supplemental/FigureA1.png")

freq2 <- freq %>%
  mutate(n = ifelse(is.na(n),0, n))

round(mean(freq2$n),2)
```

```
## [1] 39.68
```

Figure A2: Shoreline

Made in `02-StaticCovariates` in the shoreline

Figure A3: Seafloor

Made in `02-StaticCovariates` in the seafloor section

Figure A4: Sigma0 & Detection

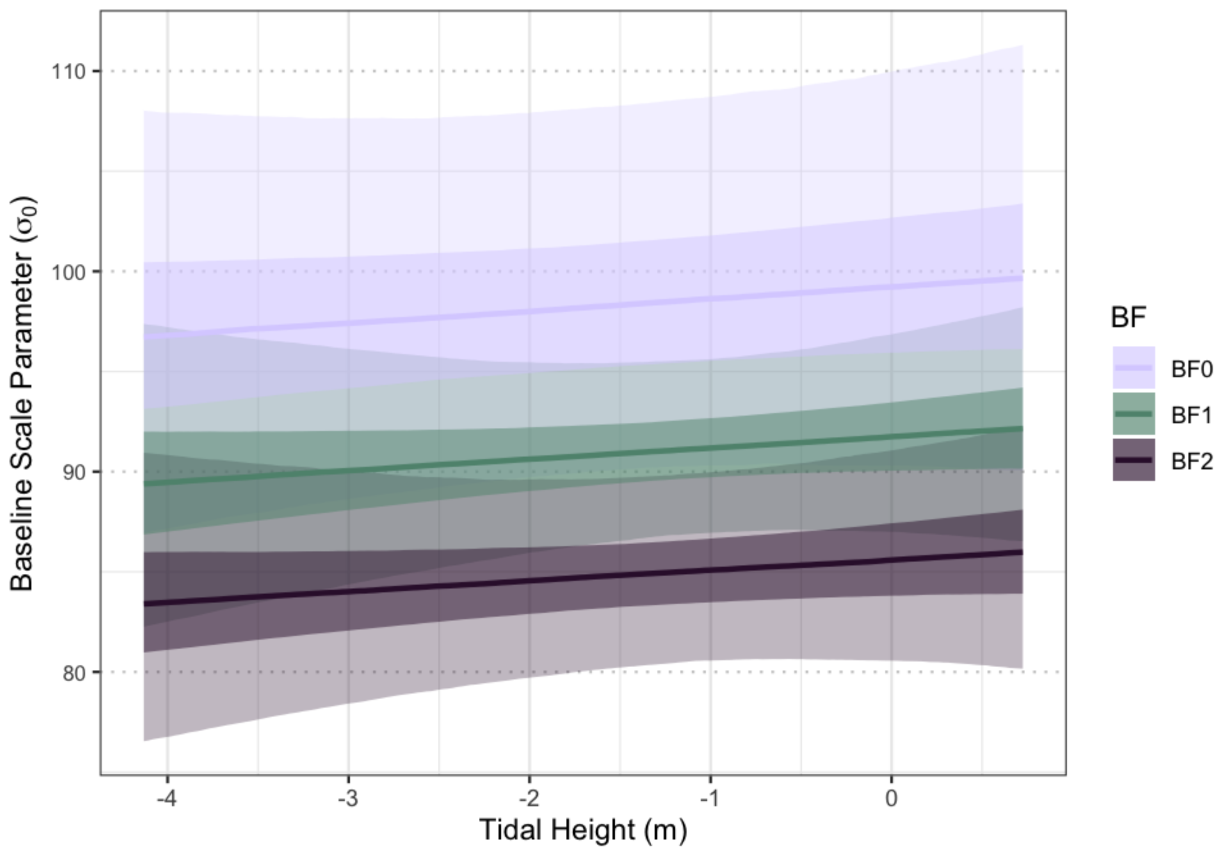
```r
range(survey_info$survey_tide, na.rm=T)
```

Tidal Heigh

```
## [1] -4.13168  0.72450
```

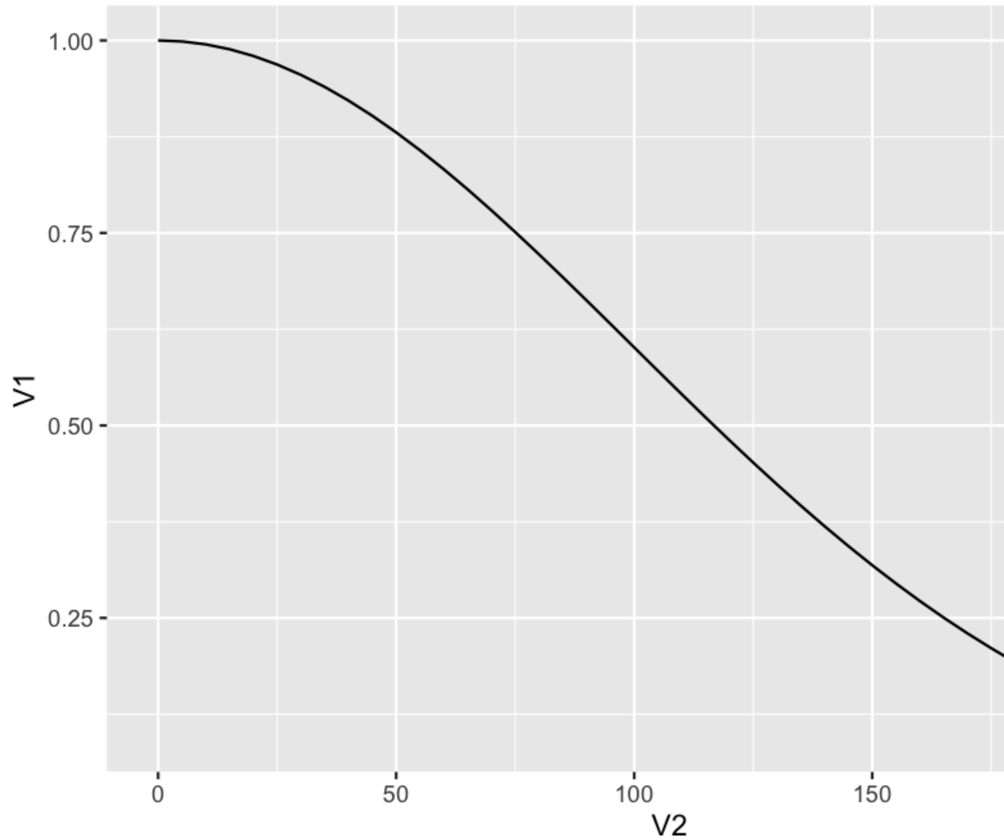
```
tide_vals <- seq(min(survey_info$survey_tide, na.rm=T),
                 max(survey_info$survey_tide, na.rm=T),length.out = 200)

# standardize
tidal_vals_sim <- (tide_vals - mean(survey_info$survey_tide, na.rm=T))/sd(survey_info$survey_tide, na.r

# get parameters
beta_tide <- MCMCchains(chains, params = "beta")

sig0 <- MCMCchains(chains, params = "sigma0")

# Pre-allocate a matrix
## BF0
sigma0 <- matrix(NA, nrow = length(beta_tide), ncol = length(tidal_vals_sim))

for (i in 1:length(beta_tide)){
  sigma0[i,] <- exp(sig0[i,1] + beta_tide[i]*tidal_vals_sim)
}

## BF1
sigma1 <- matrix(NA, nrow = length(beta_tide), ncol = length(tidal_vals_sim))

for (i in 1:length(beta_tide)){
  sigma1[i,] <- exp(sig0[i,2] + beta_tide[i]*tidal_vals_sim)
}

## BF2
sigma2 <- matrix(NA, nrow = length(beta_tide), ncol = length(tidal_vals_sim))

for (i in 1:length(beta_tide)){
  sigma2[i,] <- exp(sig0[i,3] + beta_tide[i]*tidal_vals_sim)
}

pred_tide <- data.frame(
  BF ="BF0",
  tide_scale = tidal_vals_sim,
  tide = tide_vals,
  mean = apply(sigma0, 2, mean),
  median = apply(sigma0, 2, median),
  q2 = apply(sigma0, 2, quantile, probs = 0.025),
  q97 = apply(sigma0, 2, quantile, probs = 0.975),
  q25   = apply(sigma0, 2, quantile, probs = 0.25),
  q75   = apply(sigma0, 2, quantile, probs = 0.75)) %>%
  rbind(data.frame(
    BF ="BF1",
    tide_scale = tidal_vals_sim,
    tide = tide_vals,
    mean = apply(sigma1, 2, mean),
    median = apply(sigma1, 2, median),
    q2 = apply(sigma1, 2, quantile, probs = 0.025),
```

```
    q97 = apply(sigma1, 2, quantile, probs = 0.975),
    q25   = apply(sigma1, 2, quantile, probs = 0.25),
    q75   = apply(sigma1, 2, quantile, probs = 0.75))) %>%
  rbind(data.frame(
    BF ="BF2",
    tide_scale = tidal_vals_sim,
    tide = tide_vals,
    mean = apply(sigma2, 2, mean),
    median = apply(sigma2, 2, median),
    q2 = apply(sigma2, 2, quantile, probs = 0.025),
    q97 = apply(sigma2, 2, quantile, probs = 0.975),
    q25   = apply(sigma2, 2, quantile, probs = 0.25),
    q75   = apply(sigma2, 2, quantile, probs = 0.75)))




ggplot(data= pred_tide,
       aes(y = median, x = tide, color =BF))+
  geom_ribbon(aes(ymin = q2, ymax = q97,fill=BF), color=NA, alpha=0.3)+
  geom_ribbon(aes(ymin = q25, ymax = q75, fill=BF), color=NA,alpha=0.5)+
  geom_line(aes(color =BF), linewidth = 1) +
  scale_fill_manual(values = c("#DAD2FF", "#5F927D", "#351338")) +
  scale_color_manual(values = c("#DAD2FF", "#5F927D", "#351338")) +

  ylab(expression("Baseline Scale Parameter ("*sigma[0]*")"))+
  xlab("Tidal Height (m)") +
  theme_bw()+
  theme(
      axis.text.y = element_text(size = 8),
      panel.grid.major.y = element_line(color = "grey80",
                                        linetype = "dotted"))
```



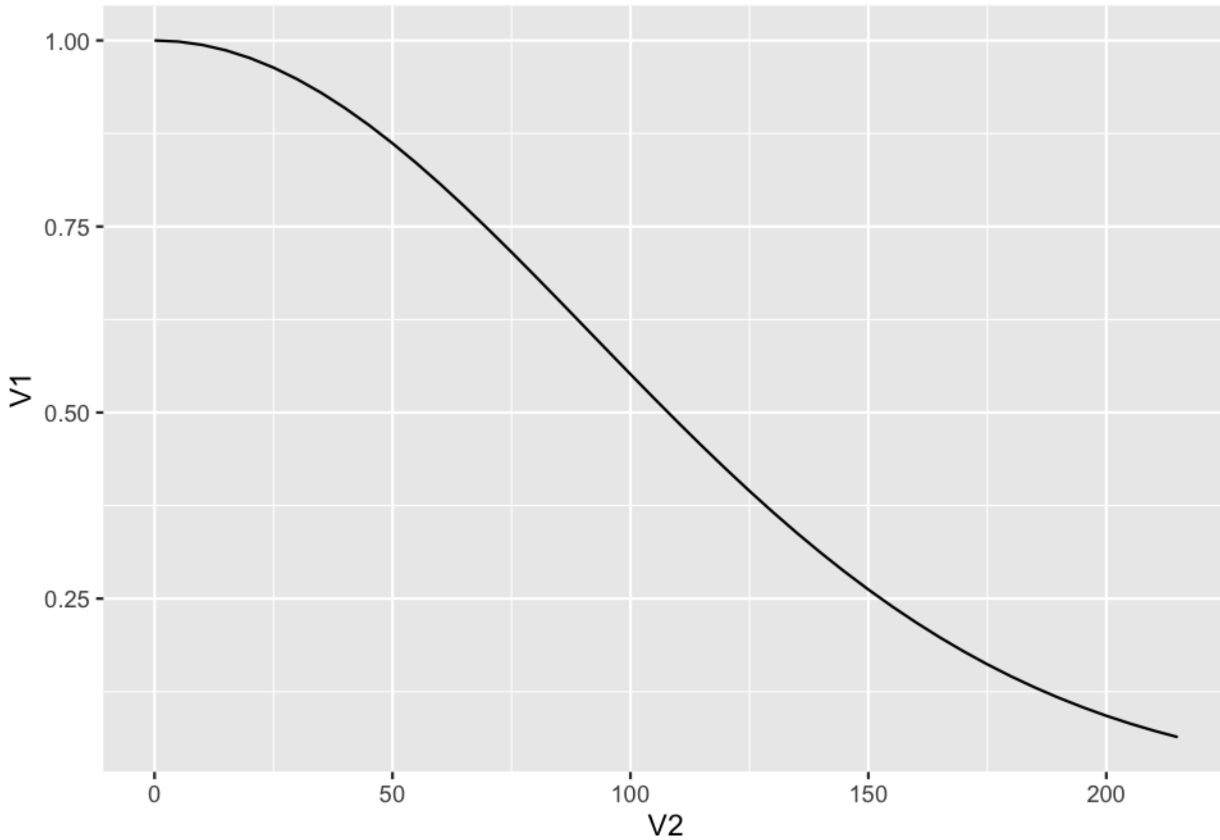

```
#ggsave("RESULTS/Supplemental/FigureA4Left.png")
```

```
dists <- seq(0, 215, 5)

distfuncHN0 <- matrix(NA, nrow = length(dists), ncol = 2)
# halfnormal detection at each distance

sigbf0_med <- median(exp(sig0[,1] + beta_tide))

distfuncHN0[, 1] <- exp(-(dists^2) / (2 * sigbf0_med^2))
distfuncHN0[, 2] <- dists


ggplot(as.data.frame(distfuncHN0), aes(x = V2, y = V1))+
  geom_line()
```

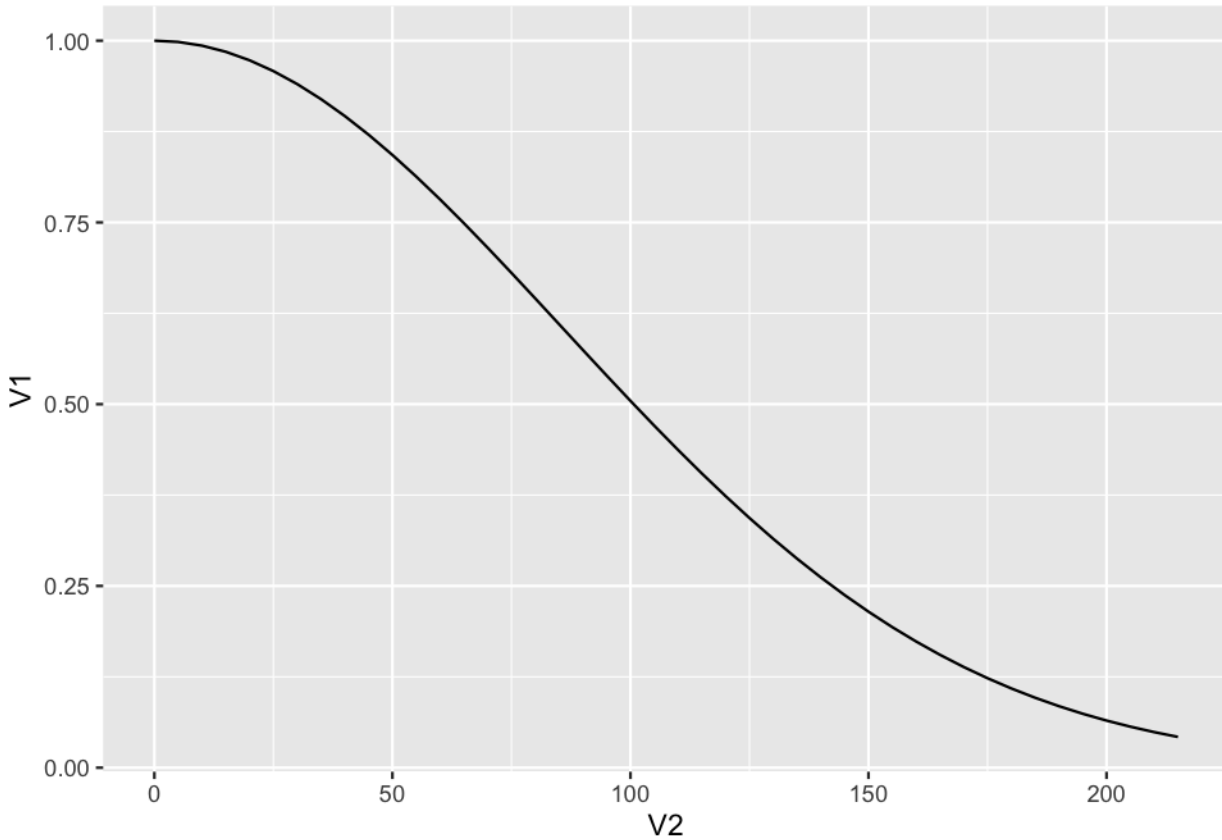


Detection Probability Distance

```
distfuncHN1 <- matrix(NA, nrow = length(dists), ncol = 2)
# Halfnormal

sigbf1_med <- median(exp(sig0[,2] + beta_tide))

distfuncHN1[, 1] <- exp(-(dists^2) / (2 * sigbf1_med^2))
distfuncHN1[, 2] <- dists


ggplot(as.data.frame(distfuncHN1), aes(x = V2, y = V1))+
  geom_line()
```

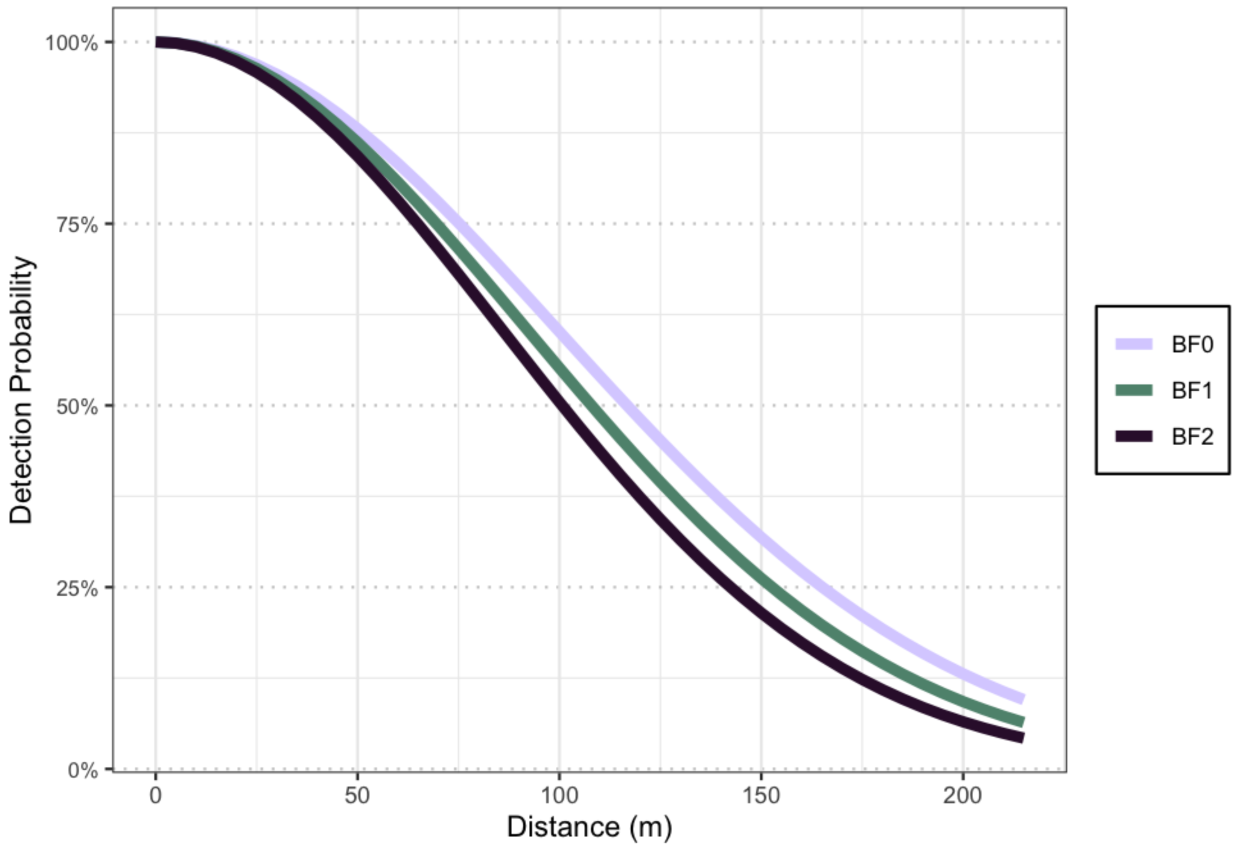



```
distfuncHN2 <- matrix(NA, nrow = length(dists), ncol = 2)
# Halfnormal

sigbf2_med <- median(exp(sig0[,3] + beta_tide))

distfuncHN2[, 1] <- exp(-(dists^2) / (2 * sigbf2_med^2))
distfuncHN2[, 2] <- dists


ggplot(as.data.frame(distfuncHN2), aes(x = V2, y = V1))+
  geom_line()
```




```
#scico(4, palette = 'managua')


detect_prob <- distfuncHN0 %>%
  as.data.frame() %>%
  mutate(Group = "BF0") %>%
  rename(pcap = V1, distance = V2) %>%
  rbind(as.data.frame(distfuncHN1) %>%
  mutate(Group = "BF1") %>%
  rename(pcap = V1, distance = V2)) %>%
  rbind(as.data.frame(distfuncHN2) %>%
  mutate(Group = "BF2")%>%
  rename(pcap = V1, distance = V2))



ggplot(detect_prob, aes(x = distance, y = pcap, color = Group)) +
  geom_line(linewidth = 2) +
  scale_color_manual(values = c("#DAD2FF", "#5F927D", "#351338")) +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  xlab("Distance (m)") +
  ylab("Detection Probability") +
  theme_bw() +
  theme(
    axis.text.y = element_text(size = 8),
    panel.grid.major.y = element_line(color = "grey80", linetype = "dotted"),
```

```r
    legend.title = element_blank(),
    legend.background = element_rect(fill = "white", color = "black"),
    legend.position.inside = c(0.8, 0.75))
```




```r
#ggsave("RESULTS/Supplemental/FigureA4Right.png")
```

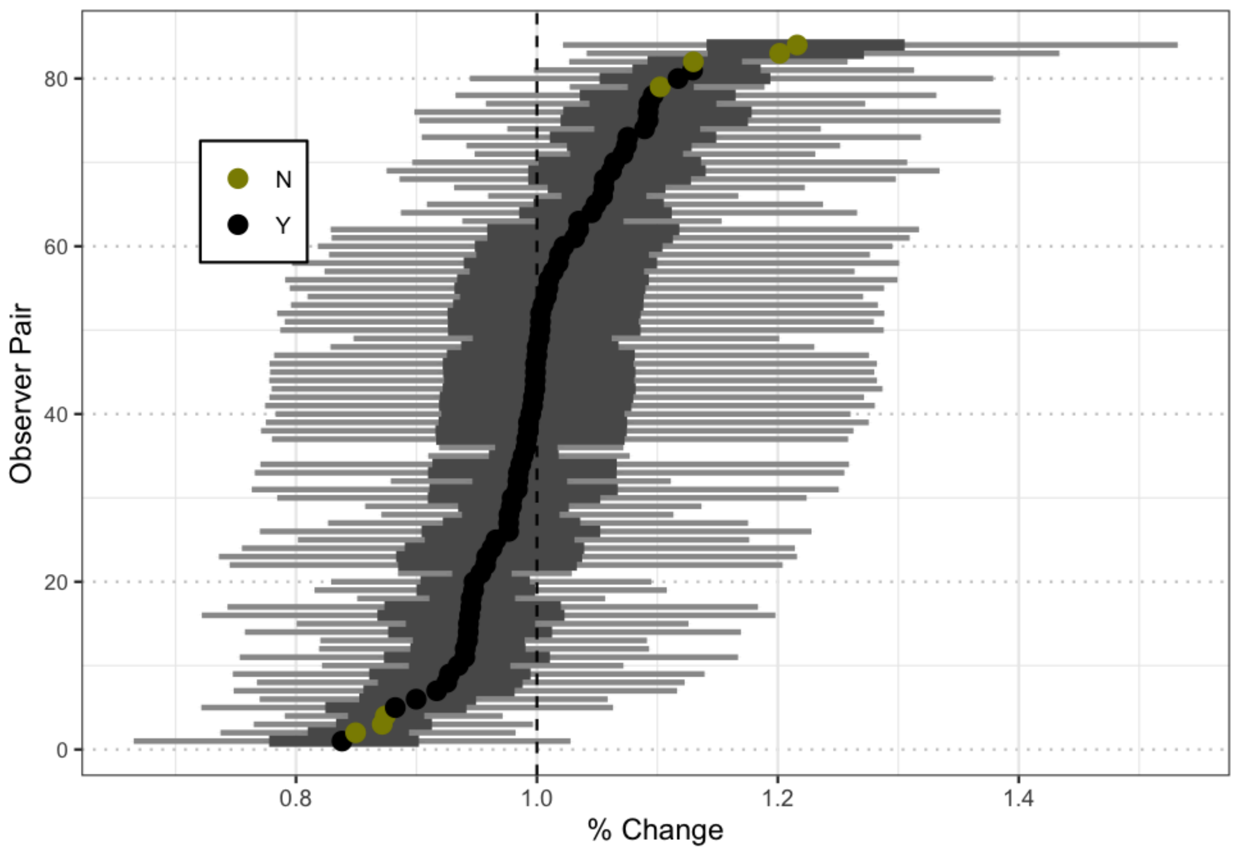
Figure A5: Observer Pairs

```r
eps_pairs <- result_summary %>%
  filter(startsWith(parameter, "eps_pair")) %>%
  mutate(overlap0 = ifelse(`2.5%` <0 & `97.5%` >0, "Y", "N")) %>%
  mutate(across(where(is.numeric), ~ exp(.x))) %>%
  arrange(`50%`) %>%
  mutate(parameter2 = as.numeric(factor(parameter, levels = parameter)))

range(eps_pairs$`50%`)
```

```
## [1] 0.8381424 1.2160482
```

```r
ggplot(data= eps_pairs,
       aes(y = parameter2, x = `50%`))+
  geom_errorbar(aes(xmin = `2.5%`, xmax = `97.5%`),
```

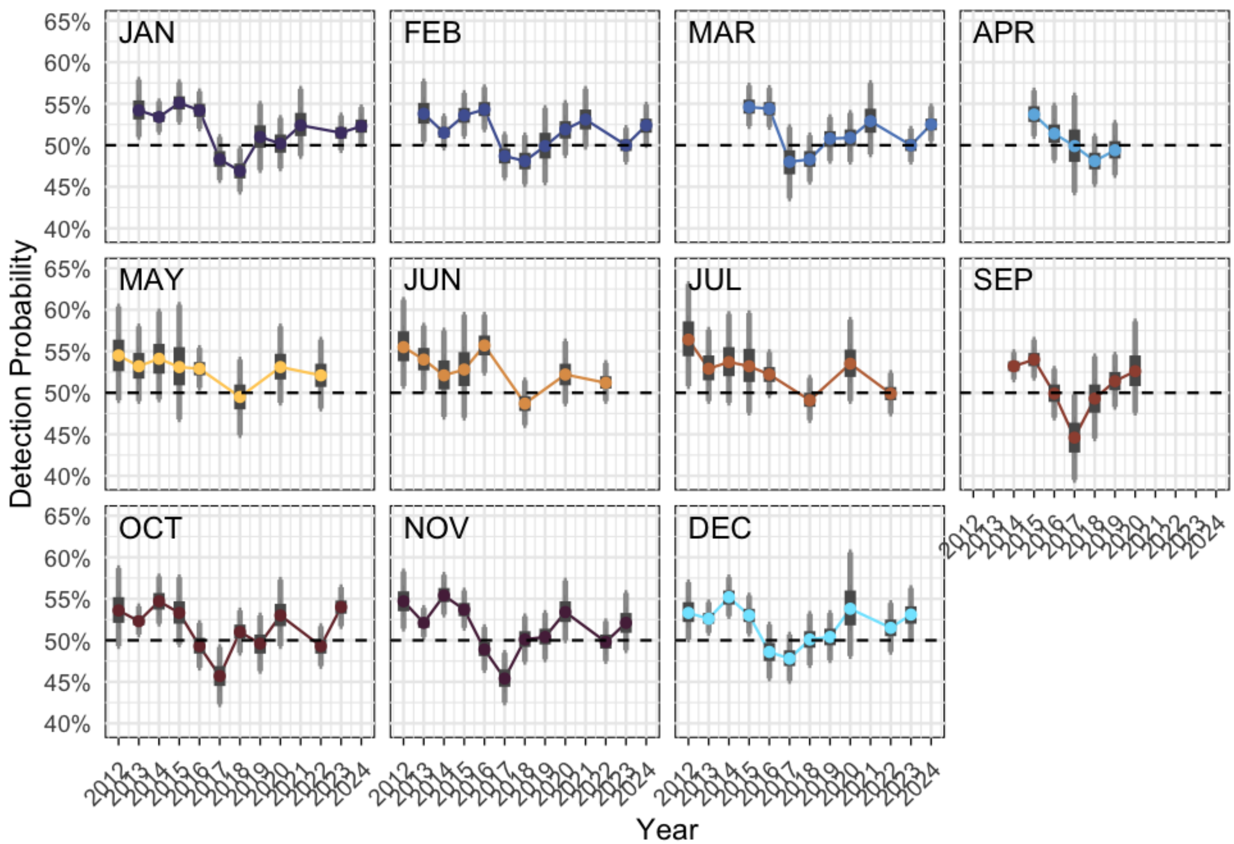
```r
                  width = 0.01, linewidth= 1,color = "grey60")+
  geom_errorbar(aes(xmin = `25%`, xmax = `75%`),
                  width = 0.01, linewidth= 2, color = "grey35")+
  geom_point(data = eps_pairs,
         aes(y = parameter2, x = `50%`, color =overlap0),size = 3) +
  theme_minimal() +
  scale_color_manual(values =c("yellow4","black"))+
  geom_vline(xintercept = 1, linetype = "dashed", color = "black") +
  xlab("% Change") +
  ylab("Observer Pair")+
  theme_bw()+
  theme(
      axis.text.y = element_text(size = 8),
      panel.grid.major.y = element_line(color = "grey80",
                                        linetype = "dotted"),
    legend.background = element_rect(fill = "white", color = "black"),
    legend.position = c(0.15, 0.75),
      legend.title=element_blank())
```




```r
#ggsave("RESULTS/Supplemental/FigureA5.png")
```

Figure A6: Pcap Monthly

```r
pcap <- result_est %>%
  filter(startsWith(parameter, "PCAP")) %>%
  mutate(monyear = str_sub(parameter, 7,14),
         mon_year_dat = format(parse_date_time(monyear, orders = "b-Y"),
                               "%m-%d-%Y"),
         mon = str_sub(monyear,1,3),
         date = as.Date(mon_year_dat, format = "%m-%d-%Y"),
         year = year(date)) %>%
  mutate(mon = factor(mon, levels= c("JAN","FEB",
                                     "MAR","APR",
                                     "MAY","JUN",
                                     "JUL","SEP",
                                     "OCT","NOV","DEC")),
         season = ifelse(mon == "MAY"| mon == "JUN" |
                           mon == "JUL", "Breeding",
                         "Non Breeding"))


# Create label positions (e.g., top-left of each facet)
label_data <- pcap %>%
  group_by(mon) %>%
  reframe(
    x = min(year),
    y = .65)

ggplot(pcap, aes(x = year, y = `50%`, color = mon)) +
  geom_line() +
  geom_errorbar(aes(ymin = `2.5%`, ymax = `97.5%`), width = 0.15,
                linewidth = 1, color = "grey60") +
  geom_errorbar(aes(ymin = `25%`, ymax = `75%`), width = 0.01,
                linewidth = 2, color = "grey35") +
  geom_point() +
  geom_text(data = label_data, aes(x = 2012, y = y, label = mon),
            inherit.aes = FALSE, hjust = 0, vjust = 1, size = 4) +
  facet_wrap(~mon, scales = "fixed") +
  scale_color_manual(values = c(
    "#4C3D72", "#5062A1", "#5F88C3", "#6EB5E1",
    "#FFCE66", "#DF9D56", "#C07348",
    "#9F503E", "#773339", "#572948","#80E6FF")) +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  scale_x_continuous(breaks = 2012:2024) +
  geom_hline(yintercept = 0.5, linetype = "dashed", color = "black") +
  labs(x = "Year", y = "Detection Probability") +
  theme_minimal() +
  theme(
    axis.ticks.x = element_line(color = "black", linewidth = 0.3),
    legend.title = element_blank(),
    legend.position = "none",
    panel.background = element_rect(color = "black"),
    axis.text.x = element_text(angle = 45, hjust = 1),
    strip.text = element_blank())
```

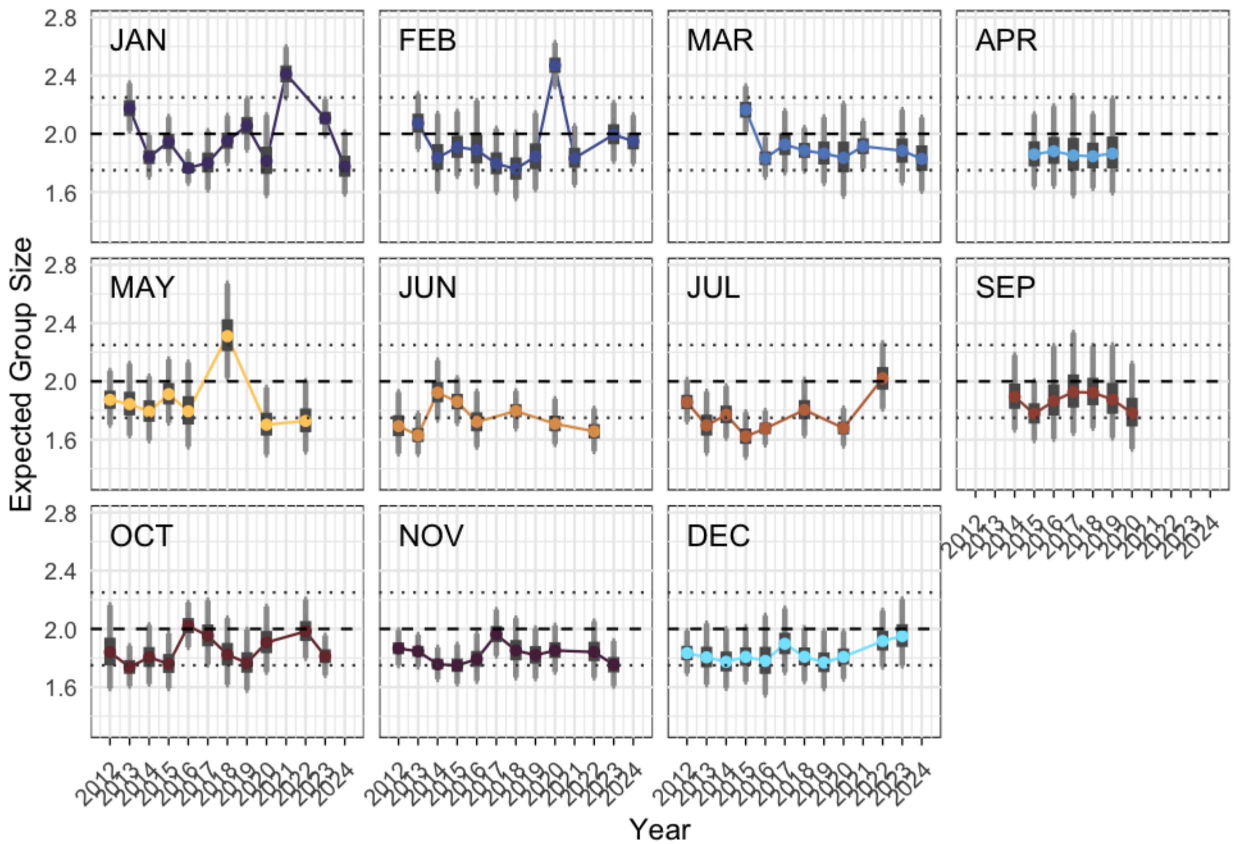



```
#ggsave("RESULTS/Supplemental/FigureA6.png",width = 10, height = 6, units = "in")
```

Figure A7: Group Sizes Monthly

```
y_bot <- 1.25
y_top <- 2.85
pad   <- 0.03 * (y_top - y_bot)

gs <- result_est %>%
  filter(startsWith(parameter, "GS")) %>%
  mutate(monyear = str_sub(parameter, 5,12),
         mon_year_dat = format(parse_date_time(monyear,
                                               orders = "b-Y"),
                               "%m-%d-%Y"),
         mon = str_sub(monyear,1,3)) %>%
  mutate(mon = factor(mon, levels= c("JAN","FEB",
                                     "MAR","APR",
                                     "MAY","JUN",
                                     "JUL","SEP",
                                     "OCT","NOV","DEC")),
         season = ifelse(mon == "MAY"|
                           mon == "JUN" | mon == "JUL",
                         "Breeding", "Non Breeding"),
         date = as.Date(mon_year_dat, format = "%m-%d-%Y"),
```

```r
         year = year(date),
         ymin_outer = pmax(`2.5%`, y_bot),
         ymax_outer = pmin(`97.5%`, y_top),
         ymin_inner = pmax(`25%`,  y_bot),
         ymax_inner = pmin(`75%`,  y_top),
         clipped_out = `97.5%` > y_top,
         label_y = y_top - pad)


label_data <- gs %>%
  group_by(mon) %>%
  reframe(x = min(as.Date(mon_year_dat, format = "%m-%d-%Y")),
    y = 2.8)

ggplot(gs, aes(x = year, y = `50%`, color = mon)) +
  geom_line() +
  geom_errorbar(aes(ymin = ymin_outer, ymax = ymax_outer),
                width = 0.15, linewidth = 1, color = "grey60") +
  geom_errorbar(aes(ymin = ymin_inner, ymax = ymax_inner),
                width = 0.01, linewidth = 2, color = "grey35") +
  geom_point() +
  geom_segment(data = subset(gs, clipped_out),
    aes(x = year, xend = year, y = y_top - 0.05, yend = y_top),
    inherit.aes = FALSE, linewidth = 0.5, color = "grey35",
    arrow = arrow(length = unit(0.15, "cm"), type = "closed")) +
  geom_text(data = label_data, aes(x = 2012,
                                   y = y, label = mon),
            inherit.aes = FALSE, hjust = 0, vjust = 1.5, size = 4) +
  facet_wrap(~ mon, scales = "fixed") +
  scale_color_manual(values = c(
    "#4C3D72", "#5062A1", "#5F88C3", "#6EB5E1",
    "#FFCE66", "#DF9D56", "#C07348",
    "#9F503E", "#773339", "#572948","#80E6FF")) +
  coord_cartesian(ylim = c(y_bot, y_top), expand = FALSE) +
  scale_x_continuous(breaks = 2012:2024, limits=c(2011,2025)) +
  geom_hline(yintercept = 2, linetype = "dashed", color = "black") +
  geom_hline(yintercept = 1.75, linetype = "dotted", color = "gray25") +
  geom_hline(yintercept = 2.25, linetype = "dotted", color = "gray25") +
  labs(x = "Year", y = "Expected Group Size") +
  theme_minimal() +
  theme(axis.ticks.x = element_line(color = "black", linewidth = 0.3),
    legend.title = element_blank(),
    legend.position = "none",
    panel.background = element_rect(color = "black"),
    axis.text.x = element_text(angle = 45, hjust = 1),
    strip.text = element_blank())
```



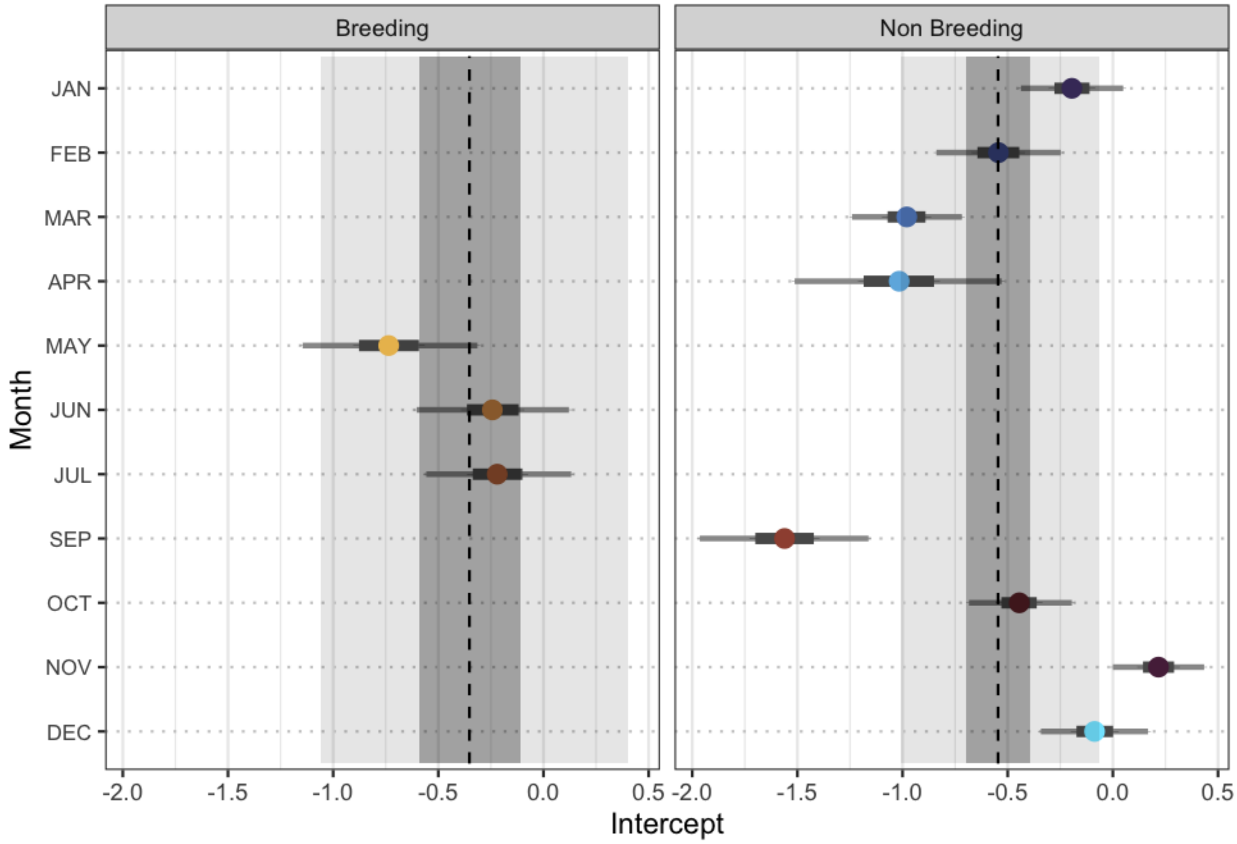

```
#ggsave("RESULTS/Supplemental/FigureA7.png", width = 10, height = 6, units = "in")
```

Figure A8: Month Random effects

```
eps_mon <- result_coef %>%
  filter(startsWith(parameter, "EPS-") &
           (endsWith(parameter,"JAN") |
           endsWith(parameter,"FEB") |
           endsWith(parameter,"MAR") |
           endsWith(parameter,"APR") |
           endsWith(parameter,"MAY") |
           endsWith(parameter,"JUN") |
           endsWith(parameter,"JUL") |
           endsWith(parameter,"SEP") |
           endsWith(parameter,"OCT") |
           endsWith(parameter,"NOV") |
           endsWith(parameter,"DEC"))) %>%
  mutate(
  season = ifelse(endsWith(parameter, "MAY")|
                    endsWith(parameter, "JUN")|
                    endsWith(parameter, "JUL"), "Breeding",
                  "Non Breeding")) %>%
  mutate(mon = str_sub(parameter, 5,7)) %>%
  mutate(mon = factor(mon, levels =c("JAN","FEB",
```

```
                                        "MAR","APR",
                                        "MAY","JUN",
                                        "JUL","SEP",
                                        "OCT","NOV",
                                        "DEC")))


alphaB <- result_coef$`50%`[result_coef$parameter=="ALPHA0 B"]
alphaNB <- result_coef$`50%`[result_coef$parameter=="ALPHA0 NB"]

intercepts <- result_coef %>%
  filter(variable == "alpha0" ) %>%
  mutate(y_min = 0.5,y_max = 11 + 0.5,
         season = ifelse(parameter == "ALPHA0 B", "Breeding", "Non Breeding"))


ggplot(data= eps_mon,
       aes(y = reorder(mon, desc(mon)), x = `50%`, color=mon))+
  geom_errorbar(aes(xmin = `2.5%`, xmax = `97.5%`),
                width = 0.01, linewidth= 1,color = "grey60")+
  geom_errorbar(aes(xmin = `25%`, xmax = `75%`),
                width = 0.01, linewidth= 2, color = "grey35")+
  geom_point(data = eps_mon,
         aes(y = reorder(mon, desc(mon)), x = `50%`),size = 3) +
  facet_wrap(~season)+
  theme_minimal() +
  scale_color_manual(values = c(
    "#4C3D72", "#5062A1", "#5F88C3", "#6EB5E1",
    "#FFCE66", "#DF9D56", "#C07348",
    "#9F503E", "#773339", "#572948","#80E6FF")) +
 geom_rect(data = intercepts,
          aes(xmin = `2.5%`, xmax = `97.5%`,
              ymin = y_min, ymax = y_max),
          inherit.aes = FALSE,
          fill = "black", alpha = 0.1) +
   geom_rect(data = intercepts,
          aes(xmin = `25%`, xmax = `75%`,
              ymin = y_min, ymax = y_max),
          inherit.aes = FALSE,
          fill = "black", alpha = 0.3) +
geom_segment(data = intercepts,
             aes(x = `50%`, xend = `50%`,
                 y = y_min, yend = y_max),
             inherit.aes = FALSE,
             linetype = "dashed", color = "black") +
  xlab("Intercept") +
  ylab("Month")+
  theme_bw()+
  theme(
      axis.text.y = element_text(size = 8),
      panel.grid.major.y = element_line(color = "grey80",
                                  linetype = "dotted"),
      legend.position = "none",
```

```r
      legend.title=element_blank())
```




```r
#ggsave("RESULTS/Supplemental/FigureA8.png")
```

Figure A9: Quadratic Distance to Shore

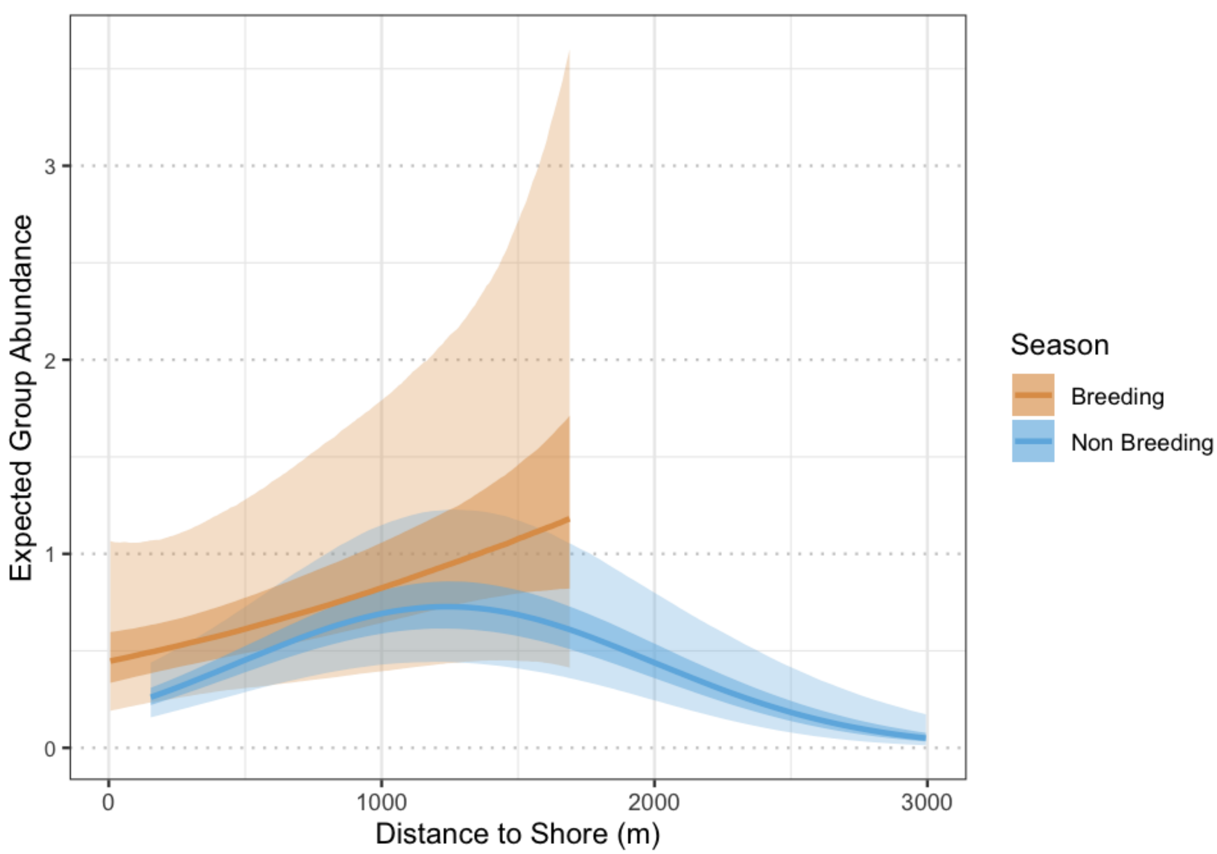
```r
range(survey_info$dist_shore, na.rm=T)
```

```
## [1]    6.000 2994.404
```

```r
shore_vals_b <- seq(min(survey_info$dist_shore[survey_info$seasNum==1],
                        na.rm=T),
                    max(survey_info$dist_shore[survey_info$seasNum==1],
                        na.rm=T),
                    length.out = 200)

# standardize
shoreB_vals_sim <- (shore_vals_b - mean(survey_info$dist_shore, na.rm=T))/
  sd(survey_info$dist_shore, na.rm=T)

shoreB_vals_simSQ <- shoreB_vals_sim*shoreB_vals_sim
```

```r
# get parameters
alpha0 <- MCMCchains(chains, params = "alpha0")

dshore <- MCMCchains(chains, params = c("alpha_sea[1, 5]","alpha_sea[1, 6]"),
                     exact=T,ISB=F)

# Pre-allocate a matrix
## Breeding
lambda_B<- matrix(NA, nrow = nrow(dshore), ncol = length(shoreB_vals_sim))

for (i in 1:nrow(dshore)){
  lambda_B[i,] <- exp(alpha0[i,1] + dshore[i,1]*shoreB_vals_sim +
                      dshore[i,2]*shoreB_vals_simSQ)
}

## Non-breeding
shore_vals_nb <- seq(min(survey_info$dist_shore[survey_info$seasNum==2],
                         na.rm=T),
                     max(survey_info$dist_shore[survey_info$seasNum==2],
                         na.rm=T),
                     length.out = 200)

# standardize
shoreNB_vals_sim <- (shore_vals_nb - mean(survey_info$dist_shore, na.rm=T))/
  sd(survey_info$dist_shore, na.rm=T)

shoreNB_vals_simSQ <- shoreNB_vals_sim*shoreNB_vals_sim

dshore <- MCMCchains(chains, params = c("alpha_sea[2, 5]","alpha_sea[2, 6]"),
                     exact=T,ISB=F)

lambda_NB<- matrix(NA, nrow = nrow(dshore), ncol = length(shoreNB_vals_sim))

for (i in 1:nrow(dshore)){
  lambda_NB[i,] <- exp(alpha0[i,2] + dshore[i,1]*shoreNB_vals_sim +
                       dshore[i,2]*shoreNB_vals_simSQ)
}


pred_shore <- data.frame(
  Season ="Breeding",
  shore_scale = shoreB_vals_sim,
  dist_shore = shore_vals_b ,
  mean = apply(lambda_B, 2, mean),
  median = apply(lambda_B, 2, median),
  q2 = apply(lambda_B, 2, quantile, probs = 0.025),
  q97 = apply(lambda_B, 2, quantile, probs = 0.975),
  q25   = apply(lambda_B, 2, quantile, probs = 0.25),
  q75   = apply(lambda_B, 2, quantile, probs = 0.75)) %>%
  rbind(data.frame(
  Season ="Non Breeding",
  shore_scale = shoreNB_vals_sim,
  dist_shore = shore_vals_nb ,
```

```
  mean = apply(lambda_NB, 2, mean),
  median = apply(lambda_NB, 2, median),
  q2 = apply(lambda_NB, 2, quantile, probs = 0.025),
  q97 = apply(lambda_NB, 2, quantile, probs = 0.975),
  q25   = apply(lambda_NB, 2, quantile, probs = 0.25),
  q75   = apply(lambda_NB, 2, quantile, probs = 0.75)))

ggplot(data= pred_shore,
       aes(y = median, x = dist_shore, color =Season))+
  geom_ribbon(aes(ymin = q2, ymax = q97,fill=Season), color=NA, alpha=0.3)+
  geom_ribbon(aes(ymin = q25, ymax = q75, fill=Season), color=NA,alpha=0.5)+
  geom_line(aes(color =Season), linewidth = 1) +
  scale_fill_manual(values = c("#DF9D56","#6EB5E1")) +
  scale_color_manual(values = c("#DF9D56","#6EB5E1")) +

  ylab("Expected Group Abundance")+
  xlab("Distance to Shore (m)") +
  theme_bw()+
  theme(
      axis.text.y = element_text(size = 8),
      panel.grid.major.y = element_line(color = "grey80", linetype = "dotted"))
```

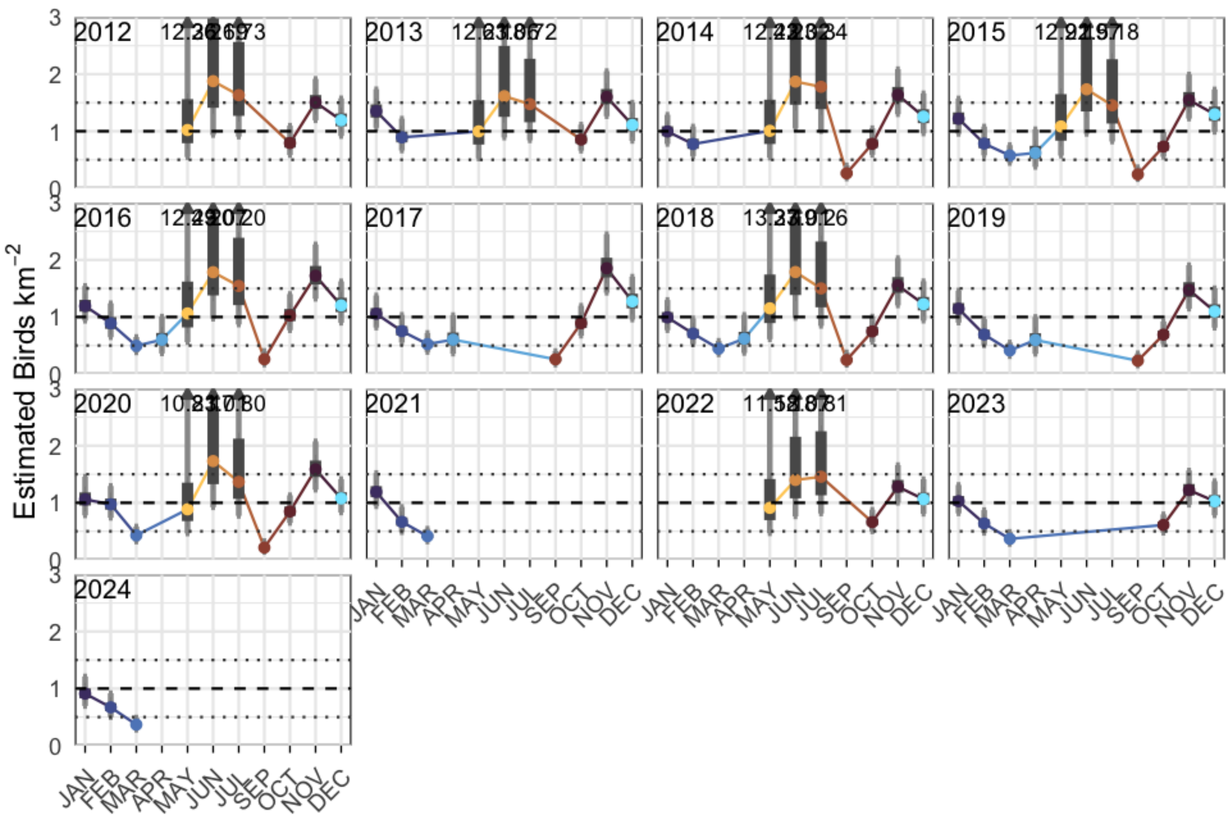



```
#ggsave("RESULTS/Supplemental/FigureA9.png")
```

Figure A10: Plot Median Density

```
all_AV_summaries <- readRDS("RESULTS/predicted_month_year_global.rds")


y_bot <- 0
y_top <- 3
pad   <- 0.03 * (y_top - y_bot)

all_AV_summaries <- all_AV_summaries %>%
  mutate(month = factor(month, levels =c("JAN","FEB","MAR","APR",
                                          "MAY","JUN","JUL",
                                          "SEP","OCT","NOV","DEC")),
         year = as.numeric(as.character(year))) %>%
  mutate(ymin_outer = pmax(LCI, y_bot),
         ymax_outer = pmin(UCI, y_top),
         ymin_inner = pmax(Q25,  y_bot),
         ymax_inner = pmin(Q75,  y_top),
         clipped_out = UCI > y_top,
         label_y = y_top - pad)

label_data2 <- all_AV_summaries %>%
  group_by(as.character(year)) %>%
  group_by(year) %>%
  summarise(x = "JAN", y = 3, .groups = "drop")




ggplot(all_AV_summaries, aes(x = month, y = MED, color = month)) +
  geom_line(group=1) +
  geom_errorbar(aes(ymin = ymin_outer, ymax = ymax_outer),
                width = 0.15, linewidth = 1, color = "grey60") +
  geom_errorbar(aes(ymin = ymin_inner, ymax = ymax_inner),
                width = 0.01, linewidth = 2, color = "grey35") +
  geom_point() +
  geom_segment(data = subset(all_AV_summaries, clipped_out),
    aes(x = month, xend = month, y = y_top - 0.05, yend = y_top),
    inherit.aes = FALSE, linewidth = 0.5, color = "grey35",
    arrow = arrow(length = unit(0.15, "cm"), type = "closed")) +
  geom_text(
  data = subset(all_AV_summaries, clipped_out),
  aes(x = month, y = label_y,
      label = paste0(scales::number(UCI, accuracy = 0.01))),
  inherit.aes = FALSE,
  vjust = 1, size = 3,
  position = position_jitter(width = 0.3, height = 0)) +

  geom_text(data = label_data2, aes(x = x, y = y, label = year),
            inherit.aes = FALSE, hjust = 0.2, vjust = 1.2, size = 3.5) +
  facet_wrap(~ year, scales = "fixed") +
  scale_color_manual(values = c(
    "#4C3D72", "#5062A1", "#5F88C3", "#6EB5E1",
```

```r
    "#FFCE66", "#DF9D56", "#C07348",
    "#9F503E", "#773339", "#572948","#80E6FF")) +
  coord_cartesian(ylim = c(y_bot, y_top), expand = FALSE,
                  xlim = c(0.6, length(unique(all_AV_summaries$month)) + 0.4)) +
  scale_x_discrete(expand = expansion(add = 0.5)) +
  scale_y_continuous(expand = expansion(mult = c(0, 0))) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "black") +
  geom_hline(yintercept = 1.5, linetype = "dotted", color = "gray25") +
  geom_hline(yintercept = 0.5, linetype = "dotted", color = "gray25") +
  labs(x = "", y = expression("Estimated Birds km"^-2)) +
  theme_minimal() +
  theme(
    axis.ticks.x = element_line(color = "black", linewidth = 0.3),
    legend.title = element_blank(),
    legend.position = "none",
    panel.background = element_rect(color = "black"),
    axis.text.x = element_text(angle = 45, hjust = 1),
    strip.text = element_blank())
```




```r
#ggsave("RESULTS/Supplemental/FigureA10.pdf",  width = 10, height = 6, units = "in")
#ggsave("RESULTS/Summplemental/FigureA10.png",  width = 10, height = 6, units = "in")
```